

Local constants of the twin-prime propagation map: explicit formulas, mean values, and an unconditional upper bound

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Abstract

Let $\mathcal{T}$ be the set of lower twin primes and, for consecutive $p_n < p_{n+1}$ in $\mathcal{T}$, let $C_n = p_n + p_{n+1} + 1$. The pair *propagates* if $C_n \in \mathcal{T}$. The Hardy–Littlewood prediction for the number of propagating pairs involves, for each gap $g = p_{n+1} - p_n$, the ratio $R(g) = \mathfrak{S}_6(g)/\mathfrak{S}_4(g)$ of the singular series of the six linear forms $n, n+2, n+g, n+g+2, 2n+g+1, 2n+g+3$ to that of the first four. We prove: (i) an explicit formula for $R(g)$ as a universal constant $R_\infty = 5.878239\dots$ times a finite product over the primes dividing $g(g^2-1)(g^2-4)(g^2-9)$, with the sharp bounds $R_{\min} = 2.083577\dots < R(g) \ll (\log \log g)^2$; (ii) the mean values $\frac{6}{G} \sum_{g \leq G, 6|g} \mathfrak{S}_4(g) \rightarrow 24C_2^2$ and $\frac{6}{G} \sum_{g \leq G, 6|g} \mathfrak{S}_6(g) \rightarrow 24C_2^2 \bar{R}$ with

$$\bar{R} = 9 \prod_{q \geq 5} \frac{q^2(q^2 - 6q + 12)}{(q-1)^2(q-2)^2} = 11.184896\dots,$$

which identifies the constant $\kappa = 2C_2\bar{R} = 14.767683\dots$ in the quantitative propagation conjecture $N(X) \sim \kappa \int_2^X dt / ((\log t)^2 (\log 2t)^2)$; (iii) the unconditional bound $N(X) \ll X/(\log X)^3$ for the number of propagating pairs with $p_{n+1} \leq X$, one logarithm short of the conjectured order; and (iv) two exact identities between singular series: bi-twin chains $(n, n+2, 2n+1, 2n+3)$ have the same singular series as prime quadruplets, and the gap-6 propagating configuration has exactly three times the singular series of a prime sextuplet. All finite computations are machine-checked and the numerical predictions are compared with data to 10^{10} , with 14,000 random twins of up to 150 digits, and with a proved orbit of the map $t \mapsto 2t + d(t) + 1$ reaching several hundred digits. Nothing here proves the twin-prime conjecture; the results are about the constants of the conjecture and an upper bound in the direction of it.

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1 Introduction and statement of results

Let $\mathcal{T} = \{3, 5, 11, 17, 29, 41, 59, 71, 101, \dots\}$ be the set of primes p with $p+2$ prime, enumerated as $p_1 < p_2 < \dots$. In [2] the author introduced the map $(p_n, p_{n+1}) \mapsto C_n := p_n + p_{n+1} + 1$ and conjectured that $C_n \in \mathcal{T}$ for infinitely many n (the *Twin-Prime Propagation Conjecture*); the companion branch $D_n = C_n + 2$ is empty for $n \geq 2$ [5]. Write

$$N(X) := \#\{n \geq 1 : p_{n+1} \leq X, C_n \in \mathcal{T}\}.$$

Both p_n and C_n are lower twins, so the natural quantitative version of the conjecture is a Hardy–Littlewood type asymptotic for $N(X)$. The purpose of this note is to determine the constants in that asymptotic exactly, and to prove what can be proved unconditionally about $N(X)$.

Notation

For a tuple of linear forms $f_i(n) = a_i n + b_i$ ($1 \leq i \leq k$) let $\nu(q)$ be the number of residues $n \bmod q$ with $q \mid f_1(n) \cdots f_k(n)$, and let

$$\mathfrak{S}(f_1, \dots, f_k) = \prod_q \left(1 - \frac{\nu(q)}{q}\right) \left(1 - \frac{1}{q}\right)^{-k} \quad (1)$$

be the singular series (the product over primes q converges absolutely when no form is a constant multiple of another). The Bateman–Horn heuristic [1] predicts $\#\{n \leq X : \text{all } f_i(n) \text{ prime}\} \sim \mathfrak{S} \cdot \int_2^X dt / \prod_i \log f_i(t)$. We write $C_2 = \prod_{q \geq 3} q(q-2)/(q-1)^2 = 0.6601618\dots$ for the twin prime constant, so that $\mathfrak{S}(n, n+2) = 2C_2$.

For an even g put

$$\mathfrak{S}_4(g) := \mathfrak{S}(n, n+2, n+g, n+g+2),$$

$$\mathfrak{S}_6(g) := \mathfrak{S}(n, n+2, n+g, n+g+2, 2n+g+1, 2n+g+3), \quad R(g) := \mathfrak{S}_6(g)/\mathfrak{S}_4(g).$$

Simultaneous primality of the six forms at n says exactly that $n, n+g \in \mathcal{T}$ and $2n+g+1 = C(n, n+g) \in \mathcal{T}$. Both series vanish unless $6 \mid g$ (Lemma 2.1), and for $6 \mid g$ the primes 2, 3 contribute the factor $4 \cdot \frac{9}{4} = 9$ to $R(g)$ (Lemma 2.2). Finally let

$$\Delta(g) := g(g^2-1)(g^2-4)(g^2-9) = (g-3)(g-2)(g-1)g(g+1)(g+2)(g+3).$$

Results

Theorem 1.1 (Explicit local ratio). *Let $6 \mid g$, $g \geq 6$. For every prime $q \geq 5$,*

$$\nu_4(g; q) = \begin{cases} 2, & q \mid g, \\ 3, & q \mid g^2 - 4, \\ 4, & \text{otherwise,} \end{cases} \quad \nu_6(g; q) = \begin{cases} \nu_4(g; q), & q \mid (g^2 - 1)(g^2 - 9), \\ \nu_4(g; q) + 2, & \text{otherwise,} \end{cases}$$

where ν_4, ν_6 are the root counts of the four- and six-form tuples. Put $\rho_q(g) = q^2(q - \nu_6)/((q-1)^2(q - \nu_4))$ and $\gamma_q = q^2(q-6)/((q-1)^2(q-4))$. Then

$$R(g) = R_\infty \rho_5(g) \rho_7(g) \prod_{\substack{q \geq 11 \\ q \mid \Delta(g)}} \frac{\rho_q(g)}{\gamma_q}, \quad R_\infty := 9 \prod_{q \geq 11} \gamma_q = 5.87823926\dots \quad (2)$$

In particular $R(g)$ is R_∞ times a finite product determined by $g \bmod 5$, $g \bmod 7$ and the prime factors ≥ 11 of $\Delta(g)$.

Corollary 1.2 (Sharp bounds). *For every $g \equiv 0 \pmod{6}$,*

$$R_{\min} := R_{\infty} \cdot \frac{25}{48} \cdot \frac{49}{72} = 2.08357729 \dots < R(g) \ll (\log \log g)^2.$$

The constant $R_{\min}$ is the infimum of $R(g)$ (approached along $g \equiv 0 \pmod{5}$, $g \equiv \pm 2 \pmod{7}$ with all prime factors of $\Delta(g)$ at least 11 large), and the upper bound is sharp up to the implied constant: $R(g) \gg (\log \log g)^2$ along $g \equiv 1 \pmod{q}$ for all $11 \leq q \leq y$, $g \leq 6 \prod_{q \leq y} q$.

Theorem 1.3 (Mean values over gaps). *As $G \rightarrow \infty$,*

$$\frac{6}{G} \sum_{\substack{g \leq G \\ 6|g}} \mathfrak{S}_4(g) \longrightarrow \frac{27}{2} \prod_{q \geq 5} \frac{q^2(q-2)^2}{(q-1)^4} = 24C_2^2 = 10.4595269 \dots, \quad (3)$$

$$\frac{6}{G} \sum_{\substack{g \leq G \\ 6|g}} \mathfrak{S}_6(g) \longrightarrow \frac{243}{2} \prod_{q \geq 5} \frac{q^4(q^2 - 6q + 12)}{(q-1)^6} = 24C_2^2 \bar{R} = 116.988726 \dots, \quad (4)$$

where

$$\bar{R} = 9 \prod_{q \geq 5} \frac{q^2(q^2 - 6q + 12)}{(q-1)^2(q-2)^2} = 11.18489648 \dots \quad (5)$$

Consequently the $\mathfrak{S}_4$ -weighted mean of $R(g)$ over gaps tends to $\bar{R}$, and

$$\kappa := 2C_2 \bar{R} = \frac{27}{2} \prod_{q \geq 5} \frac{q^3(q^2 - 6q + 12)}{(q-1)^4(q-2)} = 14.76768314 \dots \quad (6)$$

The constant κ is the one that appears in the quantitative form of the propagation conjecture (Section 5):

Conjecture 1.4 (Quantitative propagation). *As $X \rightarrow \infty$,*

$$N(X) \sim \sum_{p_{n+1} \leq X} \frac{R(p_{n+1} - p_n)}{\log C_n \log(C_n + 2)} \sim \kappa \int_2^X \frac{dt}{(\log t)^2 (\log 2t)^2}.$$

Both forms are compared with the data in Section 8: to 10^{10} the first predicts 612,777 propagating pairs against 615,447 observed (ratio 1.0044). The convergence of the mean of $R(g_n)$ to $\bar{R}$ is slow (the Hardy–Littlewood gap model predicts 11.32 at height 10^{10} ; the observed mean is 11.330), which is why the second form should be used only asymptotically.

Theorem 1.5 (Unconditional upper bound). $N(X) \ll X/(\log X)^3$.

The trivial bound is $N(X) \leq \#(\mathcal{T} \cap [1, X]) \ll X/(\log X)^2$ (the map $n \mapsto C_n$ is injective and $C_n < 2X$). The conjectured order is $X/(\log X)^4$.

Theorem 1.6 (Identities). (a) $\mathfrak{S}(n, n+2, 2n+1, 2n+3) = \mathfrak{S}(n, n+2, n+6, n+8) = 4.15118 \dots$: *bi-twin chains of length two are locally exactly as likely as prime quadruplets.*

(b) $\mathfrak{S}_6(6) = 3 \mathfrak{S}(n, n+4, n+6, n+10, n+12, n+16) = 51.8958 \dots$: *a propagating consecutive pair at gap 6 is locally exactly three times as likely as a prime sextuplet.*

Proposition 1.7 (Exceptional origins). *Call an even d productive for $t \in \mathcal{T}$ if $t + d \in \mathcal{T}$ and $2t + d + 1 \in \mathcal{T}$ [3]. The tuple $(d+t, d+t+2, d+2t+1, d+2t+3)$ in the variable d has a fixed prime divisor if and only if $t = 3$; in particular $t = 3$ has no productive gap and every other $t \in \mathcal{T}$ has an admissible gap tuple. If the side condition “ $d-1$ prime” of [3] is added, the resulting five-form tuple has a fixed prime divisor exactly for $t \in \{3, 5\}$, and for $t = 5$ the only productive d with $d-1$ prime is $d = 6$; with the side condition “ $d+1$ prime” the exceptional set is $\{3\}$.*

Remark 1.8 (What is and is not new). Theorems 1.1–1.6 are statements about Euler products and residue classes; their proofs are elementary and complete. Theorem 1.5 is a routine application of the Selberg sieve in the form of Halberstam–Richert [7, Thm. 5.7] combined with Theorem 1.3; the statement is new because the propagation map is new, the technique is not. None of this touches the parity barrier, and none of it implies the twin-prime conjecture. The method of Theorems 1.1 and 1.3 (root coincidences governed by cross differences, and averaging of singular series over a shift parameter) goes back to Hardy–Littlewood and Gallagher [6]; the specific tuples, constants and identities do not appear in the literature as far as the author has been able to determine.

2 Residues

Lemma 2.1. *If $p \in \mathcal{T}$, $p > 3$, then $p \equiv 5 \pmod{6}$. If $p < p'$ are in $\mathcal{T}$, both > 3 , then $6 \mid p' - p$. For $g \not\equiv 0 \pmod{6}$ the four forms $n, n+2, n+g, n+g+2$ have a fixed divisor 2 or 3, so $\mathfrak{S}_4(g) = \mathfrak{S}_6(g) = 0$.*

Proof. $p \equiv 1 \pmod{6}$ forces $3 \mid p+2 > 3$; $p \equiv 3 \pmod{6}$ forces $3 \mid p > 3$; even p is not prime. The difference of two residues $5 \pmod{6}$ is $0 \pmod{6}$. For odd g one of $n, n+g$ is even; for $g \equiv \pm 2 \pmod{6}$ the residues $0, -2, -g, -g-2$ cover $\mathbb{Z}/3$. $\square$

Lemma 2.2. *Let $6 \mid g$. Then $\nu_4(g; 2) = \nu_6(g; 2) = 1$ and $\nu_4(g; 3) = \nu_6(g; 3) = 2$. Hence the factors at $q = 2, 3$ of $\mathfrak{S}_4(g)$, $\mathfrak{S}_6(g)$ and $R(g)$ are $8, \frac{27}{16}, \frac{27}{2}$ and $32, \frac{243}{64}, \frac{243}{2}$ and $4, \frac{9}{4}, 9$ respectively.*

Proof. Modulo 2: n odd makes all six forms odd (since g is even), so the only root is $n \equiv 0$. Modulo 3: with $g \equiv 0$ the roots of the four forms are 0 and 1, and the two extra forms $2n+1, 2n+3 \equiv 2n+1, 2n$ vanish at $n \equiv 1$ and $n \equiv 0$, adding nothing. The factors are $(1 - \frac{1}{2})2^k = 2^{k-1}$ and $(1 - \frac{2}{3})(\frac{3}{2})^k = \frac{1}{3}(\frac{3}{2})^k$ for $k = 4, 6$. $\square$

3 The explicit local ratio

Lemma 3.1 (Root coincidences). *Let $6 \mid g$ and $q \geq 5$ prime. Modulo q the four forms $n, n+2, n+g, n+g+2$ have roots $0, -2, -g, -g-2$, and the two extra forms $2n+g+1, 2n+g+3$ have roots $r_1 = -\frac{g+1}{2}, r_2 = -\frac{g+3}{2}$.*

- (i) *Among $0, -2, -g, -g-2$ there are two coincidences if $q \mid g$, one if $q \mid g \pm 2$, none otherwise; these cases are exclusive.*
- (ii) *$r_1 \neq r_2$. Moreover $r_1 \in \{0, -2, -g, -g-2\}$ iff $q \mid (g^2 - 1)(g^2 - 9)$, and the same holds for r_2 (with the roles of $g \pm 1$ and $g \pm 3$ interchanged). Thus either both extra roots are new or both are old.*

Proof. (i) $0 \equiv -g$ or $-2 \equiv -g-2$ iff $q \mid g$; $0 \equiv -g-2$ iff $q \mid g+2$; $-2 \equiv -g$ iff $q \mid g-2$; $0 \not\equiv -2$. A prime $q \geq 5$ divides at most one of $g-2, g, g+2$.

(ii) $r_1 \equiv r_2$ iff $q \mid 2$. Solving $-\frac{g+1}{2} \equiv a$ for $a \in \{0, -2, -g, -g-2\}$ gives $q \mid g+1, g-3, g-1, g+3$ respectively; solving $-\frac{g+3}{2} \equiv a$ gives $q \mid g+3, g-1, g-3, g+1$. Both lists are the divisibility of q into one of $g \pm 1, g \pm 3$, i.e. into $(g^2 - 1)(g^2 - 9)$. Since $q \geq 5$ divides at most one of the four numbers $g-3, g-1, g+1, g+3$ (their differences are 2, 4, 6), the coincidences of r_1 and r_2 are with two distinct old roots. $\square$

Proof of Theorem 1.1. The formulas for ν_4 and ν_6 are Lemma 3.1. By Lemma 2.2 and the definition (1),

$$R(g) = 9 \prod_{q \geq 5} \frac{(1 - \nu_6/q)(1 - 1/q)^{-6}}{(1 - \nu_4/q)(1 - 1/q)^{-4}} = 9 \prod_{q \geq 5} \frac{q^2(q - \nu_6)}{(q - 1)^2(q - \nu_4)} = 9 \prod_{q \geq 5} \rho_q(g).$$

For $q \geq 11$ with $q \nmid \Delta(g)$ we have $(\nu_4, \nu_6) = (4, 6)$ and $\rho_q(g) = \gamma_q$; $\gamma_q = 1 - 9q^{-2} + O(q^{-3})$, so $\prod_{q \geq 11} \gamma_q$ converges and (2) follows on dividing out. (For $q \in \{5, 7\}$ the generic case never occurs, since $g - 3, \dots, g + 3$ are seven consecutive integers.) $\square$

Proof of Corollary 1.2. For fixed $q \geq 11$ the four possible values of ρ_q are

$$\gamma_q = \frac{q^2(q-6)}{(q-1)^2(q-4)}, \quad \frac{q^2(q-4)}{(q-1)^2(q-2)} \quad (q \mid g),$$

$$\frac{q^2(q-5)}{(q-1)^2(q-3)} \quad (q \mid g^2 - 4), \quad \frac{q^2}{(q-1)^2} \quad (q \mid (g^2 - 1)(g^2 - 9)),$$

and γ_q is the smallest: $(q-4)^2 > (q-2)(q-6)$ and $(q-5)(q-4) > (q-6)(q-3)$. Hence every factor in (2) beyond $R_\infty \rho_5 \rho_7$ is > 1 , and $R(g) \geq R_\infty \min \rho_5 \min \rho_7$ with equality only when no prime ≥ 11 divides $\Delta(g)$. Enumerating $g \bmod 5$: $\rho_5 = \frac{25}{48}$ for $g \equiv 0$ and $\frac{25}{16}$ otherwise; $g \bmod 7$: $\rho_7 = \frac{49}{60}, \frac{49}{36}, \frac{49}{72}$ for $g \equiv 0, \pm 1$ or $\pm 3, \pm 2$. So $R(g) \geq R_{\min}$. Equality would need $g \equiv 0 \pmod{5}$ and every prime factor of $\Delta(g)$ below 11; by Sylvester–Schur (a product of 7 consecutive integers all > 7 has a prime factor > 7) this forces $g \leq 10$, and $g = 6$ has $g \equiv 1 \pmod{5}$. So $R(g) > R_{\min}$ for all g , while choosing $g \equiv 0 \pmod{5}$, $g \equiv \pm 2 \pmod{7}$ with all prime factors of $\Delta(g)$ above 11 large gives $R(g) \rightarrow R_{\min}$.

Upper bound: $\rho_q/\gamma_q \leq \frac{q^2}{(q-1)^2}/\gamma_q = \frac{q-4}{q-6} = 1 + \frac{2}{q-6}$, so $\log R(g) \leq \log(R_\infty \rho_5 \rho_7) + \sum_{q \mid \Delta(g), q \geq 11} \frac{2}{q-6}$. The sum has $\omega(\Delta(g)) \leq \log \Delta(g)/\log 11 \ll \log g$ terms and is largest when the primes are the smallest possible, i.e. at most $\sum_{11 \leq q \leq y} \frac{2}{q-6}$ with $\pi(y) \ll \log g$, so $y \ll \log g \log \log g$ and the sum is $2 \log \log y + O(1) \leq 2 \log \log \log g + O(1)$. Hence $R(g) \ll (\log \log g)^2$. Conversely, for $y \geq 11$ choose by the Chinese remainder theorem $g \equiv 0 \pmod{6}$ with $g \equiv 1 \pmod{q}$ for every prime $11 \leq q \leq y$ and $0 < g \leq 6 \prod_{q \leq y} q = e^{(1+o(1))y}$. Then every such q divides $g-1 \mid \Delta(g)$ with $\rho_q/\gamma_q = 1 + \frac{2}{q-6}$, so $R(g) \geq R_\infty \rho_5 \rho_7 \prod_{11 \leq q \leq y} (1 + \frac{2}{q-6}) \gg (\log y)^2 \gg (\log \log g)^2$. $\square$

4 Mean values

Throughout this section $k \in \{4, 6\}$, $\mathfrak{S}_k(g)$ is as above and, for a prime $q \geq 5$,

$$f_q(g) := \left(1 - \frac{\nu_k(g; q)}{q}\right) \left(1 - \frac{1}{q}\right)^{-k}, \quad u_q(g) := f_q(g) - 1,$$

so that $\mathfrak{S}_k(g) = c_k \prod_{q \geq 5} f_q(g)$ with $c_4 = \frac{27}{2}$, $c_6 = \frac{243}{2}$ (Lemma 2.2). Note that $f_q(g)$ depends only on $g \bmod q$.

Lemma 4.1 (Local means). *For every prime $q \geq 5$,*

$$\frac{1}{q} \sum_{a=0}^{q-1} \left(1 - \frac{\nu_4(a; q)}{q}\right) = \frac{(q-2)^2}{q^2}, \quad \frac{1}{q} \sum_{a=0}^{q-1} \left(1 - \frac{\nu_6(a; q)}{q}\right) = \frac{q^2 - 6q + 12}{q^2}.$$

Proof. By Lemma 3.1, $\sum_a (q - \nu_4(a; q)) = (q-2) + 2(q-3) + (q-3)(q-4) = (q-2)^2$ (one class with $q \mid a$, two with $q \mid a \pm 2$, $q-3$ generic). For ν_6 : the class $a \equiv 0$ contributes $q-4$, the classes $a \equiv \pm 2$ contribute $2(q-5)$, the four classes $a \equiv \pm 1, \pm 3$ contribute $4(q-4)$, and the remaining $q-7$ classes contribute $(q-7)(q-6)$; the total is $q^2 - 6q + 12$. This decomposition assumes the seven classes $0, \pm 1, \pm 2, \pm 3$ are distinct, i.e. $q \geq 7$; for $q = 7$ there are then no generic classes and the total is $3 + 4 + 12 = 19 = 49 - 42 + 12$. For $q = 5$ the classes ± 2 coincide with ∓ 3 and have $(\nu_4, \nu_6) = (3, 3)$, so the five classes contribute $1 + 2 \cdot 2 + 2 \cdot 1 = 7 = 25 - 30 + 12$. (Both identities are also verified by enumeration for $5 \leq q \leq 200$ in the accompanying code.) $\square$

Lemma 4.2 (Size of the local deviations). *There is an absolute constant C (one may take $C = 11$) such that $|u_q(g)| \leq C/q$ for all $q \geq 5$, $k \in \{4, 6\}$ and all g . If moreover $q \nmid \Delta(g)$ and $q \geq 11$ then $|\log f_q(g)| \leq 2k^2/q^2$.*

Proof. Write $u_q = [(1 - \nu/q) - (1 - 1/q)^k]/(1 - 1/q)^k$. For $q \geq 7$ the binomial expansion of $(1 - 1/q)^k$ is alternating with decreasing terms (consecutive terms have ratio $\frac{k-j}{(j+1)q} \leq k/q < 1$), so $1 - k/q \leq (1 - 1/q)^k \leq 1 - k/q + k(k-1)/(2q^2)$ and, since $2 \leq \nu \leq k$, the numerator lies in $[-k(k-1)/(2q^2), (k-\nu)/q]$, of absolute value at most $4/q$; the denominator is at least $(6/7)^6 > 0.39$. The finitely many cases $q = 5$ are checked directly (the largest deviation is $0.53 < 11/5$). For the generic factor,

$$-\log f_q = -\log(1 - k/q) + k \log(1 - 1/q) = \sum_{j \geq 2} \frac{k^j - k}{j q^j} \in \left[0, \frac{k^2}{2q^2} \cdot \frac{1}{1 - k/q}\right],$$

and $1/(1 - k/q) \leq 11/5$ for $q \geq 11$, $k \leq 6$. $\square$

Lemma 4.3 (Averaging a periodic function along multiples of 6). *Let d be coprime to 6, let $u : \mathbb{Z} \rightarrow \mathbb{R}$ be d -periodic with $|u| \leq B$ and mean $\bar{u} = \frac{1}{d} \sum_{a=0}^{d-1} u(a)$. Then for every $M \geq 1$, $|\sum_{m=1}^M u(6m) - M\bar{u}| \leq 2dB$. If $d = q_1 \cdots q_r$ is a product of distinct primes and $u(g) = \prod_i u_{q_i}(g)$ with u_{q_i} q_i -periodic, then $\bar{u} = \prod_i \bar{u}_{q_i}$.*

Proof. Since 6 is invertible modulo d , $m \mapsto 6m$ permutes the residues modulo d , so the sum of $u(6m)$ over any d consecutive m equals $d\bar{u}$. Split $[1, M]$ into $\lfloor M/d \rfloor$ full blocks and a remainder of fewer than d terms. The second statement is the Chinese remainder theorem: $\mathbb{Z}/d \cong \prod \mathbb{Z}/q_i$ and the mean of a product of functions of the separate coordinates is the product of the means. $\square$

Proof of Theorem 1.3. Fix $k \in \{4, 6\}$, let G be large, $M = \lfloor G/6 \rfloor$ and $Q = \lfloor \log G \rfloor$. Put $P = \prod_{5 \leq q \leq Q} q$ and

$$\mathfrak{S}_k^{(Q)}(g) = c_k \prod_{5 \leq q \leq Q} f_q(g) = c_k \prod_{5 \leq q \leq Q} (1 + u_q(g)) = c_k \sum_{d|P} u_d(g), \quad u_d(g) := \prod_{q|d} u_q(g).$$

Truncation. For $6 \mid g \leq G$, $\mathfrak{S}_k(g)/\mathfrak{S}_k^{(Q)}(g) = \prod_{q > Q} f_q(g)$. By Lemma 4.2 the primes $q > Q$ not dividing $\Delta(g)$ contribute $\exp(O(\sum_{q > Q} q^{-2})) = 1 + O(1/Q)$. The primes $q > Q$ dividing $\Delta(g)$ number at most $\log |\Delta(g)|/\log Q \leq 7 \log(G+3)/\log Q$, and each contributes a factor $1 + O(1/q) = 1 + O(1/Q)$. Hence, uniformly in $g \leq G$,

$$\mathfrak{S}_k(g) = \mathfrak{S}_k^{(Q)}(g) \left(1 + O\left(\frac{\log G}{Q \log Q}\right)\right) = \mathfrak{S}_k^{(Q)}(g) \left(1 + O\left(\frac{1}{\log \log G}\right)\right). \quad (7)$$

Averaging. Each u_d is d -periodic with $|u_d| \leq \prod_{q|d} C/q =: B_d$ (Lemma 4.2), and $\gcd(d, 6) = 1$. By Lemma 4.3,

$$\sum_{m=1}^M \mathfrak{S}_k^{(Q)}(6m) = c_k \sum_{d|P} \left(M \prod_{q|d} \bar{u}_q + O(dB_d) \right) = c_k M \prod_{5 \leq q \leq Q} (1 + \bar{u}_q) + O\left(c_k \prod_{5 \leq q \leq Q} (1 + C)\right),$$

because $\sum_{d|P} dB_d = \prod_{q \leq Q} (1 + q \cdot C/q)$. The error is $(1 + C)^{\pi(Q)} \leq (1 + C)^{2 \log G / \log \log G} = G^{o(1)} = o(M)$.

Local means. By Lemma 4.1, $1 + \bar{u}_q = \frac{(q-2)^2}{q^2} \left(1 - \frac{1}{q}\right)^{-4} = \frac{q^2(q-2)^2}{(q-1)^4}$ for $k = 4$ and $1 + \bar{u}_q = \frac{q^2 - 6q + 12}{q^2} \left(1 - \frac{1}{q}\right)^{-6} = \frac{q^4(q^2 - 6q + 12)}{(q-1)^6}$ for $k = 6$; in both cases $\bar{u}_q = O(q^{-2})$, so $\prod_{q \geq 5} (1 + \bar{u}_q)$ converges and $\prod_{5 \leq q \leq Q} (1 + \bar{u}_q) = \prod_{q \geq 5} (1 + \bar{u}_q) (1 + O(1/Q))$.

Combining the three steps,

$$\frac{1}{M} \sum_{m \leq M} \mathfrak{S}_k(6m) = c_k \prod_{q \geq 5} (1 + \bar{u}_q) \left(1 + O\left(\frac{1}{\log \log G}\right)\right),$$

which is (3) and (4) with the stated Euler products. Finally $\prod_{q \geq 5} \frac{q(q-2)}{(q-1)^2} = \frac{C_2}{3/4} = \frac{4C_2}{3}$ gives $\frac{27}{2} \cdot \frac{16C_2^2}{9} = 24C_2^2$; dividing (4) by (3) gives (5), and $2C_2\bar{R}$ with $2C_2 = \frac{3}{2} \prod_{q \geq 5} \frac{q(q-2)}{(q-1)^2}$ gives (6). $\square$

Remark 4.4. The error term $O(1/\log \log G)$ is what the crude truncation gives; the partial means converge much faster in practice (relative error about $40/G$ at $G \in [6 \cdot 10^4, 1.2 \cdot 10^5]$). A Ramanujan-sum expansion in the style of Montgomery–Soundararajan would give a power saving, but is not needed here.

5 The constant of the propagation conjecture

For a consecutive pair (p_n, p_{n+1}) with gap g , simultaneous membership $p_n, p_{n+1}, C_n \in \mathcal{T}$ is primality of the six forms at $n = p_n$. The Bateman–Horn heuristic for the six forms, divided by that for the four forms (which describe the event “ p_n and $p_n + g$ are lower twins” that we condition on), gives the conditional probability $R(g)/(\log C_n \log(C_n + 2))$ that the pair propagates. Summing over pairs gives the first form of Conjecture 1.4. (Being consecutive is an additional condition on the pair; heuristically it is independent of the primality of $C_n, C_n + 2$, which live near $2p_n$, and it is imposed in both numerator and denominator of the conditional probability.)

For the second form one needs the average of $R(g_n)$ over consecutive pairs. In the Hardy–Littlewood model the gap g following a lower twin near height x has probability proportional to $\mathfrak{S}_4(g) e^{-\lambda g}$ with $\lambda = 2C_2/(\log x)^2$ the density of $\mathcal{T}$ (the factor $\mathfrak{S}_4(g)$ is the local correlation of the two twins, the exponential is the probability of no twin in between). Since $\lambda \rightarrow 0$, the average over g of $R(g)$ against this weight tends, by Theorem 1.3 and partial summation, to the $\mathfrak{S}_4$ -weighted mean $\bar{R}$. Together with $\#(\mathcal{T} \cap [1, x]) \sim 2C_2 x/(\log x)^2$ and $\log C_n \sim \log 2p_n$ this gives $N(X) \sim 2C_2 \bar{R} \int dt/((\log t)^2 (\log 2t)^2)$, i.e. $\kappa = 2C_2 \bar{R}$.

The finite-height average

$$\bar{R}(x) := \frac{\sum_{6|g} \mathfrak{S}_6(g) e^{-\lambda g}}{\sum_{6|g} \mathfrak{S}_4(g) e^{-\lambda g}}, \quad \lambda = \frac{2C_2}{(\log x)^2},$$

converges to $\bar{R}$ only like the partial means of Theorem 1.3 at $G \approx 1/\lambda$. Table 1 lists it; the observed mean of $R(g_n)$ over all consecutive pairs with $p_n \leq 10^{10}$ is 11.3297 (11.3467 to 10^9), and over the dyadic blocks $[2^{28}, 2^{29})$ and $[2^{33}, 2^{34})$ it is 11.335 and 11.320, in agreement with the model at those heights and drifting in the predicted direction.

x	$\bar{R}(x)$	mean gap $1/\lambda$ -weighted
10^6	11.448	154
10^8	11.366	268
10^9	11.340	337
10^{10}	11.319	414
10^{12}	11.290	592
10^{20}	11.238	1623
10^{50}	11.201	$\approx 10^4$
∞	11.18490	

Table 1: Model mean of $R(g_n)$ at height x and its limit $\bar{R}$.

6 The unconditional upper bound

We use the Selberg upper-bound sieve in the following form.

Theorem 6.1 (Halberstam–Richert [7, Thm. 5.7]). *Let $F_i(n) = a_i n + b_i$ ($1 \leq i \leq k$) be linear forms with integer coefficients, $a_i \geq 1$, such that $E := \prod_i a_i \prod_{r < s} (a_r b_s - a_s b_r) \neq 0$, and let*

$\rho(q)$ be the number of roots of $\prod_i F_i$ modulo q . Suppose $\rho(q) < q$ for all primes q . Then for $1 < y \leq z$,

$$\#\{z-y < n \leq z : F_1(n), \dots, F_k(n) \text{ all prime}\} \leq 2^k k! \prod_q \left(1 - \frac{\rho(q)}{q}\right) \left(1 - \frac{1}{q}\right)^{-k} \frac{y}{(\log y)^k} (1 + O(\varepsilon)),$$

where $\varepsilon = (\log \log 3y + \log \log 3|E|)/\log y$ and the implied constant depends only on k .

Lemma 6.2. For every $g \equiv 0 \pmod{6}$ the six forms $n, n+2, n+g, n+g+2, 2n+g+1, 2n+g+3$ satisfy $\nu_6(g; q) < q$ for every prime q .

Proof. $\nu_6 = 1 < 2$ and $\nu_6 = 2 < 3$ by Lemma 2.2; for $q \geq 5$, $\nu_6 \leq \nu_4 + 2 \leq 6$, which is $< q$ for $q \geq 7$. For $q = 5$ use Lemma 3.1: 5 divides exactly one of $g-2, g-1, g, g+1, g+2$. If $5 \mid g \pm 1$ then $\nu_6 = \nu_4 \leq 4$. If $5 \mid g \pm 2$ then also $5 \mid g \mp 3$ (as $g+2 \equiv g-3$), so again $\nu_6 = \nu_4 = 3$. If $5 \nmid g$ then $\nu_4 = 2$ and $\nu_6 \leq 4$. In every case $\nu_6(g; 5) \leq 4 < 5$. $\square$

Proof of Theorem 1.5. Let X be large and $G = \lfloor (\log X)^3 \rfloor$. Every n counted by $N(X)$ has $p_{n+1} \leq X$; write $g_n = p_{n+1} - p_n$.

Large gaps. The gaps g_n with $p_{n+1} \leq X$ are disjoint intervals inside $[3, X]$, so $\sum_{p_{n+1} \leq X} g_n \leq X$ and

$$\#\{n : p_{n+1} \leq X, g_n > G\} \leq X/G \ll X/(\log X)^3.$$

Small gaps. Fix $g \leq G$ with $6 \mid g$ (other g do not occur by Lemma 2.1, apart from the single pair $(3, 5)$). If $g_n = g$ and the pair propagates, then $n' := p_n \leq X$ makes the six forms $n', n'+2, n'+g, n'+g+2, 2n'+g+1, 2n'+g+3$ simultaneously prime. Apply Theorem 6.1 with $k = 6$, $y = z = X$: the hypothesis $\rho(q) < q$ is Lemma 6.2; the coefficients are $a_i \in \{1, 2\}$ and $|b_i| \leq g+3$, so $|E| \leq 2^6(2(g+3))^{15} \leq (\log X)^{60}$ and $\log \log 3|E| \ll \log \log \log X$. Hence

$$\#\{n : p_{n+1} \leq X, g_n = g, C_n \in \mathcal{T}\} \ll \mathfrak{S}_6(g) \frac{X}{(\log X)^6},$$

uniformly in $g \leq G$. Summing over $6 \mid g \leq G$ and using Theorem 1.3 (which gives $\sum_{6 \mid g \leq G} \mathfrak{S}_6(g) \ll G$),

$$\#\{n : p_{n+1} \leq X, g_n \leq G, C_n \in \mathcal{T}\} \ll \frac{X}{(\log X)^6} \sum_{\substack{g \leq G \\ 6 \mid g}} \mathfrak{S}_6(g) \ll \frac{GX}{(\log X)^6} \ll \frac{X}{(\log X)^3}.$$

Adding the two parts proves the theorem. $\square$

Remark 6.3. The loss of one logarithm against the conjectured $X/(\log X)^4$ comes entirely from the trivial treatment of large gaps: one would need $\#\{p_n \leq X : g_n > (\log X)^3\} \ll X/(\log X)^4$, a statement about twin primes in short intervals that is not available. Any improvement of the large-gap count improves Theorem 1.5 correspondingly.

7 Identities and exceptional origins

Proof of Theorem 1.6. (a) For the bi-twin tuple $n, n+2, 2n+1, 2n+3$ the roots modulo $q \geq 5$ are $0, -2, -\frac{1}{2}, -\frac{3}{2}$; their pairwise differences are $2, \frac{1}{2}, \frac{3}{2}, \frac{3}{2}, \frac{1}{2}, 1$ (up to sign), none of which vanishes modulo $q \geq 5$, so $\nu(q) = 4$. For the quadruplet $n, n+2, n+6, n+8$ the roots $0, -2, -6, -8$ have differences $2, 6, 8, 4, 6, 2$, again nonzero modulo $q \geq 5$, so $\nu(q) = 4$. Modulo 2 both tuples have $\nu = 1$; modulo 3 both have $\nu = 2$ (roots $0, 1$ in each case). The Euler products coincide factor by factor.

(b) The propagation tuple with $g = 6$ has, by Theorem 1.1, $\nu_6(6; 5) = 4$ (as $5 \mid g-1$), $\nu_6(6; 7) = 4$ (as $7 \mid g+1$), and $\nu_6(6; q) = 6$ for $q \geq 11$ because $\Delta(6) = 2^6 \cdot 3^4 \cdot 5 \cdot 7$ has no prime

factor ≥ 11 ; and $\nu = 1, 2$ at $q = 2, 3$. The sextuplet $n, n+4, n+6, n+10, n+12, n+16$ has roots $0, -4, -6, -10, -12, -16$; modulo 5 these are $\{0, 1, 4, 0, 3, 4\}$, four classes; modulo 7 they are $\{0, 3, 1, 4, 2, 5\}$, six classes; the pairwise differences $4, 6, 10, 12, 16, 2, 6, 8, 12, 4, 6, 10, 2, 6, 4$ are not divisible by any prime ≥ 11 , so $\nu = 6$ for $q \geq 11$; and $\nu = 1, 2$ at $q = 2, 3$. The two root-count profiles agree except at $q = 7$, where the local factors are $(1 - \frac{4}{7})(\frac{6}{7})^{-6}$ and $(1 - \frac{6}{7})(\frac{6}{7})^{-6}$, whose ratio is 3. $\square$

Proof of Proposition 1.7. Modulo 2 all four forms $d+t, d+t+2, d+2t+1, d+2t+3$ are odd for even d . Modulo 3, for $t \equiv 2$ (every $t \in \mathcal{T}$, $t > 3$) the roots in d are $-t \equiv 1, -t-2 \equiv 2, -2t-1 \equiv 1, -2t-3 \equiv 2$, so $d \equiv 0$ survives; for $t = 3$ the roots are $0, 1, 2, 0$, covering $\mathbb{Z}/3$. For $q \geq 5$ there are at most four roots. Hence the tuple has a fixed divisor iff $t = 3$; and for $t = 3$ any d with $3+d \in \mathcal{T}$ has $d \equiv 2 \pmod{6}$, so $2 \cdot 3 + d + 1 \equiv 0 \pmod{3}$ and exceeds 3: no productive gap exists.

With the fifth form $d-1$: modulo 5 and $t = 5$ the roots are $0, 3, 4, 2, 1$, all of $\mathbb{Z}/5$, so exactly one of $d+5, d+7, d+11, d+13, d-1$ is divisible by 5 for every d ; primality forces that one to equal 5, which happens only for $d-1 = 5$, i.e. $d = 6$ (and indeed $11, 13, 17, 19$ are prime). For $t \in \mathcal{T}$, $t \geq 11$, one has $t \bmod 30 \in \{11, 17, 29\}$ and a direct enumeration gives $\nu_5 \leq 4$ in all three cases and for both side conditions; modulo 3 the fifth form $d \mp 1 \equiv \mp 1$ adds no root at $d \equiv 0$; for $q \geq 7$ there are at most five roots. With the form $d+1$ and $t = 5$ the roots modulo 5 are $0, 3, 4, 2, 4$, so the tuple is admissible. This gives the exceptional sets $\{3, 5\}$ and $\{3\}$. $\square$

8 Numerical verification

All constants were evaluated as Euler products truncated at $Q = 10^8$ with the analytic tail $\sum_{q>Q} q^{-2} \approx (1 - 1/\log Q)/(Q \log Q)$ added; changing Q from 10^7 to 10^8 moves the values by less than $2 \cdot 10^{-9}$. As a check, the product form of $24C_2^2$ in (3) agrees with the value computed from the known digits of C_2 to 10^{-10} .

$R_\infty = 9 \prod_{q \geq 11} \gamma_q$	5.87823926
$R_{\min} = R_\infty \cdot \frac{25}{48} \cdot \frac{49}{72}$	2.08357729
$R(6) = R_\infty \cdot \frac{25}{16} \cdot \frac{49}{36}$	12.5014637
$24C_2^2$	10.4595270
$\bar{R}$	11.1848965
$24C_2^2 \bar{R}$	116.988726
$\kappa = 2C_2 \bar{R}$	14.7676831
$\mathfrak{S}(\text{bi-twin}) = \mathfrak{S}(\text{quadruplet})$	4.15118075
$\mathfrak{S}_6(6) = 3 \mathfrak{S}(\text{sextuplet})$	51.8958333

Theorem 1.1 was checked against the direct product over the same primes for g up to 9,699,690 (relative difference $< 10^{-12}$), and the root-count table against enumeration for all $6 \mid g \leq 3000$, $5 \leq q \leq 300$. The minimum of $R(g)$ over $6 \mid g \leq 2 \cdot 10^5$ is 2.083876 at $g = 195,540$ (so $R(g) > R_{\min}$ with the infimum approached), and the maximum is 42.51 at $g = 194,466$. Lemma 4.1 was checked by enumeration for $5 \leq q \leq 200$, and the partial means of Theorem 1.3 to $G = 1.2 \cdot 10^5$ are 10.4560 and 116.970 against the limits 10.4595 and 116.989.

The propagation conjecture to 10^{10}

Among the 27,412,679 lower twins up to 10^{10} there are 615,447 propagating consecutive pairs; the first form of Conjecture 1.4 predicts 612,777.0 (ratio 1.0044) and the κ -integral 612,694.8 (ratio 1.0045). The per-dyadic-block ratios are within $\pm 1.7\%$ from 2^{22} on and within $\pm 0.7\%$ from 2^{28} on. The only pair with $D_n \in \mathcal{T}$ is $(3, 5)$, as in [5]. Restricting to gap 6: there are

180,529 consecutive pairs at gap 6 with $p_n \leq 10^{10}$ (prime quadruplets) against the Hardy–Littlewood prediction 181,065.1, and 4,527 of them propagate against the Bateman–Horn prediction $\mathfrak{S}_6(6) \int_2^{10^{10}} dt / ((\log t)^2 (\log(t+6))^2 (\log(2t+7))^2) = 4,540.6$ for the six forms (ratio 0.997), with $\mathfrak{S}_6(6) = 3 \mathfrak{S}(\text{sextuplet})$ from Theorem 1.6. Per gap, the ratios observed/predicted for the twenty most frequent gaps $g \leq 120$ lie between 0.945 and 1.036.

The minimal productive gap

For $t \in \mathcal{T}$, $t \geq 5$, let $d(t)$ be the least $d \equiv 0 \pmod{6}$ that is productive for t in the sense of [3] (i.e. $t + d \in \mathcal{T}$, $2t + d + 1 \in \mathcal{T}$, and $d - 1$ or $d + 1$ prime). The note [3] conjectures that $d(t)$ exists for every $t \geq 5$; the Bateman–Horn heuristic for the five-form tuple in the variable d , with the fixed divisors of Proposition 1.7 removed, gives an expected number $E_t(Y)$ of productive gaps $\leq Y$ of order $Y/(\log t)^4$, and the Poisson model $\Pr(d(t) > Y) = e^{-E_t(Y)}$ then predicts $d(t) \asymp (\log t)^4 \log \log t$ on average and $d(t) \leq (\log t)^5 \log \log t$ for every $t \geq 5$ (the refined form of the conjecture used in the repository). Three independent data sets test this.

Exhaustive range. Every one of the 27,412,678 lower twins $5 \leq t \leq 10^{10}$ has a productive gap; the largest is $d(9,432,106,511) = 1,728,906$, and the largest value of $d(t)/((\log t)^5 \log \log t)$ is 0.144, at $t = 324,301,421$. The mean of $d(t)/((\log t)^4 \log \log t)$ is 0.0573 or 0.0574 in every dyadic block from $[2^{26}, 2^{27})$ to $[2^{33}, 2^{34})$, and the model mean $\sum_Y e^{-E_t(Y)}$ matches the observed block means to within 10% throughout and to 0.1% in the last block. The side condition is met by $d - 1$ alone for 43.3% of the twins, by $d + 1$ alone for 42.9%, and by both for 13.8%.

Independent samples at large height. For each of the sizes 20, 30, 40, 60, 80, 100, 150 decimal digits, 2,000 lower twins were drawn uniformly at random (random start in the decade, forward scan) and $d(t)$ was computed exactly by sieving the candidates $d = 6m$ with all primes up to 10^5 before testing survivors. Under the Poisson model the values $u = e^{-E_t(d(t))}$ are uniform on $(0, 1)$. Over the 14,000 samples the mean of u is 0.5033 and the Kolmogorov–Smirnov distance to the uniform distribution is 0.0086, against a 5% critical value of 0.0115; the seven per-size distances are 0.018, 0.039, 0.023, 0.019, 0.028, 0.017, 0.020 (critical value 0.030), and the ratios $\sum d(t) / \sum \mathbb{E}[d(t)]$ lie between 0.954 and 1.022. The normalised mean $d(t)/((\log t)^4 \log \log t)$ is 0.056–0.061 at every size, the same value as in the exhaustive range.

The constructive orbit. Iterating $t \mapsto 2t + d(t) + 1$ from $t = 5$ produces a single orbit through $\mathcal{T}$ (the “twin-gap chain” of [3]). It was followed for 1,134 steps, to a t of 343 decimal digits. Every prime along it (t , $t + 2$, $t + d$, $t + d + 2$ at each step, and the final pair) was proved prime with the APR–CL test of PARI/GP, and ECPP certificates for the final pair are distributed with the repository, so the orbit does not rest on probable-prime tests. Along the orbit the values $u = e^{-E_t(d(t))}$ have mean 0.516 and Kolmogorov–Smirnov distance 0.038 to the uniform distribution (5% critical value 0.040); the largest value of $d(t)/((\log t)^5 \log \log t)$ is 0.556, attained at the first step $t = 5$, and it is below 0.005 for every $t > 10^{12}$. The orbit therefore behaves, as far as $d(t)$ is concerned, like a sequence of independent twins of the same sizes, across some 340 orders of magnitude.

All checks are reproduced in the accompanying repository¹ by the commands `certify`, `theorems`, `propagation` and `gaps` of the `twinconj` package and by the scripts `extend_chain`, `sample_gaps` and `certify_chain`; the two exhaustive runs to 10^{10} need about 40 GB of memory.

¹<https://github.com/dacomb-bierton/Local-Constants-Twin-Prime-Propogation>

9 Status

Claim	Status
Explicit formula and sharp bounds for $R(g)$	Theorem 1.1, Corollary 1.2: proved.
Mean values of $\mathfrak{S}_4, \mathfrak{S}_6$ over gaps; closed forms for $\overline{R}, \kappa$ $N(X) \ll X/(\log X)^3$	Theorem 1.3: proved. Theorem 1.5: proved (Selberg sieve).
Bi-twin = quadruplet, $\mathfrak{S}_6(6) = 3 \mathfrak{S}(\text{sextuplet})$	Theorem 1.6: proved.
Exceptional origins $\{3\}, \{3, 5\}$	Proposition 1.7: proved.
$N(X) \rightarrow \infty$ (propagation conjecture)	Open; follows from Hypothesis H [5].
$N(X) \sim \kappa \int dt/((\log t)^2(\log 2t)^2)$	Conjecture 1.4; consistent with data to 10^{10} (ratio 1.0044).
Twin-prime conjecture	Open; not addressed by anything here.

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